VP150 Midterm RC

Section 1 & 2

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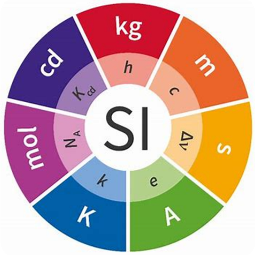
Units

Important in dimensional analysis and vector calculation.

Factor	Name	Symbol	Factor	Name	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^{9}	giga	G	10^{-12}	pico	p
10^{6}	mega	M	10^{-15}	femto	f
10^{3}	kilo	k	10^{-18}	atto	a
10^{2}	hecto	h	10^{-21}	zepto	z
10^{1}	deka	da	10^{-24}	yocto	y



Scalar & Vector Quantity

- Scalar: mass m, distance $|\overrightarrow{d}|$, speed $|\overrightarrow{v}|$...
- Vector: weight $\overrightarrow{W} = m\overrightarrow{g}$, displacement $\overrightarrow{d}$, velocity $\overrightarrow{v}$, force, linear momentum, angular velocity, angular momentum, electric/magnetic field, electric current density, ...

Vector quantities have both a **magnitude** and a **direction**. Be careful with that and state them clearly in your answer.

Vector Operation

- Addition & Subtraction: parallelogram rule
- Multiplication by a scalar
- Scalar (Dot) Product:

$$\overrightarrow{A} \cdot \overrightarrow{B} = |\overrightarrow{A}||\overrightarrow{B}| \cos\alpha$$

- **Vector (Cross) Product** Magnitude:
  $|\overrightarrow{A} \times \overrightarrow{B}| = |\overrightarrow{A}||\overrightarrow{B}| \sin\alpha = \text{Area enclosed by the parallelogram}$

  Direction: right hand rule

Important Property

- Prove parallelism

$$\overrightarrow{A} \times \overrightarrow{B} = 0 \Longleftrightarrow \overrightarrow{A} \parallel \overrightarrow{B}$$

- Prove perpendicularity

$$\overrightarrow{A} \cdot \overrightarrow{B} = 0 \Longleftrightarrow \overrightarrow{A} \perp \overrightarrow{B}$$

Vector Operation in Cartesian Coordinate System

$$\begin{aligned}\overrightarrow{A} &= A_x \overrightarrow{i} + A_y \overrightarrow{j} + A_z \overrightarrow{k} \\ \overrightarrow{B} &= B_x \overrightarrow{i} + B_y \overrightarrow{j} + B_z \overrightarrow{k}\end{aligned}$$

Scalar (Dot) Product

$$\overrightarrow{A} \cdot \overrightarrow{B} = A_x B_x + A_y B_y + A_z B_z$$

Vector (Cross) Product

$$\begin{aligned}\overrightarrow{A} \times \overrightarrow{B} &= \begin{pmatrix} \overrightarrow{i} & \overrightarrow{j} & \overrightarrow{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{pmatrix} \\ &= (A_y B_z - A_z B_y)\overrightarrow{i} + (\mathbf{A_z B_x} - \mathbf{A_x B_z})\overrightarrow{j} + (A_x B_y - A_y B_x)\overrightarrow{k}\end{aligned}$$

Exercise 1

Let $\vec{u} = 3\vec{n}_x - \vec{n}_y + \vec{n}_z$ and $\vec{w} = \vec{n}_x - 3\vec{n}_y + 2\vec{n}_z$. What is the magnitude of the component of the vector $2\vec{u} - 3\vec{w}$ pointing in the direction of the vector $\vec{u} \times \vec{w}$?

Solution

What is the property of $\vec{u} \times \vec{w}$?

$$(\vec{u} \times \vec{w}) \perp \vec{u}$$

$$(\vec{u} \times \vec{w}) \perp \vec{w}$$

What about a linear combination of $\vec{u}$ and $\vec{w}$?

$$(\vec{u} \times \vec{w}) \perp (2\vec{u} - 3\vec{w})$$

The magnitude of component: 0

Section 1 - Homework 1

A rectangle piece of Aluminum is 5.10 ± 0.01 cm long and $1.90 \pm 0.01 cm$ wide.
a) Find the area of the rectangle and the uncertainty in the area
b) Verify that the fractional uncertainty in the area is equal to the sum of the fractional uncertainties in the length and in the width.

Solution

a) First, we use the extreme values in the pieces length and width to find the uncertainty in the area:

$5.11 \times 1.91 - 5.1 \times 1.9$

The length could be as large as 5.11 cm and the width could be as large as 1.91 cm. The area is $9.69 \pm 0.07 cm^2$.

The fractional uncertainty in the area is $0.07/9.69 = 0.72\%$

b) the fractional uncertainties in the length and width are:

$$0.01/5.11 = 0.20\% \text{ and } 0.01/1.9 = 0.53\%.$$

The sum of these fractional uncertainties is:

$$0.20\% + 0.53\% = 0.73\%,$$

in agreement with the fractional uncertainty in the area.

Section 1 - Homework 1

The drag force F on a sphere is related to the radius of the sphere, r, the velocity of the sphere, v, and the coefficient of viscosity of the fluid the drop is falling through, η, by the formula:

$$F = kr^x \eta^y v^z$$

where k is a dimensionless constant, and x, y, and z are integers. By considering the units of the equation, work out the values of x, y, and z. (The coefficient of viscosity η has units of $kg \cdot m^{-1} \cdot s^{-1}$).

Solution

The units must be the same on both sides of the equation. Given that k is dimensionless:

$$F = kr^x \eta^y v^z$$

$$kg \cdot m \cdot s^{-2} = m^x \cdot (kg \cdot m^{-1} \cdot s^{-1})^y \cdot (m \cdot s^{-1})^z$$

It Is obvious that $y = 1$ considering kg (only one term).
Re-arranging the various terms gives:

$$m \cdot s^{-2} = m^{x+z-1} \cdot m^{-y} \cdot s^{-z-1}$$

From which it is straightforward obtaining $x = 1$ and $z = 1$. The final equation is simply:

$$F = kr\eta v$$

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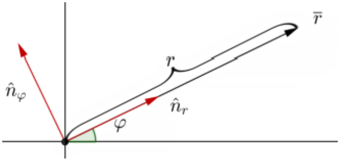
Polar Coordinate System

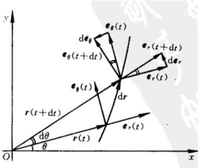
From Cartesian to Polar:

$$\vec{e}_r = \cos\theta\,\vec{i} + \sin\theta\,\vec{j}$$
$$\vec{e}_\theta = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}$$



Polar Coordinate System

Make sure you understand how these formulas are derived.



$$\frac{\mathrm{d}}{\mathrm{d}t}\overrightarrow{e}_r = \frac{\mathrm{d}\theta}{\mathrm{d}t}\overrightarrow{e}_\theta$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\overrightarrow{e}_\theta = -\frac{\mathrm{d}\theta}{\mathrm{d}t}\overrightarrow{e}_r$$

$$\vec{v} = \frac{\mathrm{d}r}{\mathrm{d}t}\vec{e_r} + r\frac{\mathrm{d}\theta}{\mathrm{d}t}\vec{e_\theta}$$

$$\vec{a} = \underbrace{(\frac{\mathrm{d}r^2}{\mathrm{d}^2t} - r(\frac{\mathrm{d}\theta}{\mathrm{d}t})^2)\vec{e_r}}_{radial} + \underbrace{(2\frac{\mathrm{d}r}{\mathrm{d}t}\frac{\mathrm{d}\theta}{\mathrm{d}t} + r\frac{\mathrm{d}\theta^2}{\mathrm{d}^2t})\vec{e_\theta}}_{transverse}$$

Component of v & a in Cartesian & Polar Coordinate

- **Tangential** component: tangent to the direction of trajectory(with respect to trajectory)

$$\vec{v}_T = v(t)\vec{e}_T$$
$$\vec{a}_T = \frac{\mathrm{d}}{\mathrm{d}t}v(t)\vec{e}_T$$

- **Normal**: perpendicular to tangential component

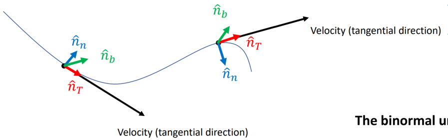
$$\vec{a}_N = \sqrt{a^2 - a_T^2}\,\vec{e}_N$$

- **Radial** component: in radial direction, pointing from the origin to current position(with respect to origin)
- **Transverse** component: perpendicular to radial component

Natural Coordinate System

- Tangential
$$\vec{e}_T = \frac{\vec{v}}{|\vec{v}|}$$
- Normal
$$\vec{e}_N = \frac{\frac{\mathrm{d}}{\mathrm{d}t}\vec{e}_T}{\left|\frac{\mathrm{d}}{\mathrm{d}t}\vec{e}_T\right|}$$
- Binormal
$$\vec{n}_B = \vec{e}_T \times \vec{e}_N$$

Natural (kinematic) coordinate system



$\hat{n}_T$ tangential
$\hat{n}_n$ normal
$\hat{n}_b$ binormal

The binormal unit vector:

$$\hat{n}_b = \hat{n}_T \times \hat{n}_n$$

Right-hand rule applies

Component of v & a in Natural Coordinate

$$
\begin{aligned}
\vec{v} &= v(t)\vec{e}_T \\
\vec{a} &= \frac{\mathrm{d}}{\mathrm{d}t}v(t)\vec{e}_T + v(t)\left|\frac{\mathrm{d}\vec{e}_T}{\mathrm{d}t}\right|\vec{e}_N \\
&= \frac{\mathrm{d}}{\mathrm{d}t}v(t)\vec{e}_T + \frac{v^2(t)}{R_c}\vec{e}_N
\end{aligned}
$$

where **Radius of Curvature**

$$
R_c = \frac{v}{\left|\frac{\mathrm{d}\vec{e}_T}{\mathrm{d}t}\right|}
$$

1D-Kinematics: Average & Instantaneous Velocity

Since in a 1D system, the position can be determined by a function $x(t)$, the **average velocity** within Δt can be calculated:

$$v_{av} = \frac{x(t_0 + \Delta t) - x(t_0)}{\Delta t}.$$

When Δt goes to 0, the **instantaneous velocity** at t_0 can be derived:

$$v(t_0) = \lim_{\Delta t \to 0} \frac{x(t_0 + \Delta t) - x(t_0)}{\Delta t} = \dot{x}(t_0).$$

Generally, the corresponding velocity function can be expressed as:

$$v_x(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} = \dot{x}(t).$$

1D-Kinematics: Average & Instantaneous Acceleration

Since velocity represents the rate that the position changes, and acceleration represents the rate that velocity changes, the relationship between acceleration and velocity can be derived similarly.

$$a_{av} = \frac{v(t_0 + \Delta t) - v(t_0)}{\Delta t}$$

$$a_x(t) = \frac{\mathrm{d}v(t)}{\mathrm{d}t} = \dot{v}(t) = \ddot{r}(t)$$

1D-Kinematics: Distance and Speed

- **Instantaneous speed** is the magnitude of the instantaneous velocity.
- **Distance**

$$d = \int_{t_0}^{t_1} |v(t)| \, \mathrm{d}t.$$

2D &3 D-Kinematics

- **Displacement**

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

- **Velocity**

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\vec{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\vec{j} + \frac{\mathrm{d}z}{\mathrm{d}t}\vec{k} = (\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t})$$

- **Acceleration**

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}v_x}{\mathrm{d}t}\vec{i} + \frac{\mathrm{d}v_y}{\mathrm{d}t}\vec{j} + \frac{\mathrm{d}v_z}{\mathrm{d}t}\vec{k} = (\frac{\mathrm{d}v_x}{\mathrm{d}t}, \frac{\mathrm{d}v_y}{\mathrm{d}t}, \frac{\mathrm{d}v_z}{\mathrm{d}t})$$

Application to Circular Motion

- Uniform Circular Motion:

$$\vec{v} = \frac{\mathrm{d}r}{\mathrm{d}t}\vec{e_r} + r\frac{\mathrm{d}\theta}{\mathrm{d}t}\vec{e_\theta} = r\omega\vec{e_\theta}$$

$$\vec{a} = \frac{\mathrm{d}}{\mathrm{d}t}\vec{v} = -r\omega^2\vec{e_r} = -\frac{v^2}{r}\vec{e_r}$$

- Non-Uniform Circular Motion:

$$\vec{v}(t) = r\omega(t)\vec{e_\theta}$$

$$\vec{a}(t) = -r\omega^2(t)\vec{e_r} + r\epsilon(t)\vec{e_\theta}$$

where $\quad \omega(t) = \frac{\mathrm{d}}{\mathrm{d}t}\theta(t) \qquad \epsilon(t) = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\theta(t)$

Application to Projectile Motion

$$v_x(t) = v_0 \cos\alpha$$
$$v_y(t) = v_0 \sin\alpha - gt$$
$$y(t) = v_0 \sin\alpha t - \frac{1}{2}gt^2$$
$$x(t) = v_0 \cos\alpha t$$
$$H = y_{max} = \frac{v_0^2 \sin^2\alpha}{2g}$$
$$L = x_{max} = \frac{v_0^2 \sin 2\alpha}{g}$$

when $\alpha = 45^\circ$

$$L_{max} = \frac{v_0^2}{g}$$

Exercises 1

A balloon takes off the ground and moves so that the vertical component of its velocity remains constant and equal to v_0. Because of the wind, the balloon drifts to the east with horizontal velocity $v_x = by$, where y is the distance from the ground (i.e. the altitude) and b is a positive constant.

- Find the position of the balloon at any instant of time.
- Find the equation of the balloon trajectory as $y = y(x)$ or $x = x(y)$ and sketch the curve.
- Determine the total acceleration of the balloon and the magnitudes of its tangential and normal components.

Solution

1. State coordinate $t=0$, balloon's position as origin, use cartisian.

horizon: x-axis. to sky: y-axis

$V_y = V_0 \rightarrow y = V_0 t$

$V_x = by \rightarrow x = byt = bV_0t^2$

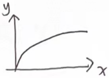
$$\boxed{\vec{r} = bV_0t^2\,\hat{n}_x + V_0 t\,\hat{n}_y}$$

2. $y = V_0 t \rightarrow t = \frac{y}{V_0}$

$x = byt$

$$\boxed{x = \frac{b}{2V_0}y^2} \quad \boxed{y \geq 0}$$

DOMAIN!



3. $\cancel{a_x}$

$V_x = bV_0t \qquad a_x = bV_0$

$V_y = V_0 \qquad a_y = 0$

$$\boxed{\vec{a} = bV_0\,\hat{n}_x}$$

in natural coordinate. $\vec{V} = v(t)\,\hat{n}_t = V_0\sqrt{1+b^2t^2}\,\hat{n}_t$.

$\vec{a} = \dot{v}(t)\,\hat{n}_t + \frac{v^2(t)}{R}\,\hat{n}_n$ hard to calculate.

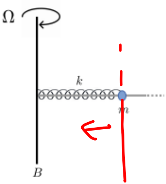
$a_t = \left(V_0\sqrt{1+b^2t^2}\right)' = \frac{b^2V_0t}{\sqrt{1+b^2t^2}}$

$a_n = \sqrt{|a|^2 - a_t^2} = \frac{bV_0}{\sqrt{1+b^2t^2}}$

Exercise 2

$\Omega = \sqrt{k/m}$

Question 4. (*7 points*) [*Note.* There is no gravitational force in this question.] A linear spring with spring constant k and unstretched/uncompressed length is x_0 is coiled around a light long horizontal rod (see the figure). The rod is fixed to a vertical axle B, which spins always with the same angular speed $\Omega = \sqrt{k/m}$. One end of the spring is fixed at the axle. At the other end of the spring there is bead with mass m that can slide along the rod, but the rod is rough so that the bead moves with friction.



What should the value of the coefficient of kinetic friction μ be, so that the bead slides away from the axle with constant speed v_0? Comment briefly on whether the scenario described in this problem is sustainable?

Solution

use polar coordinate., B as origin, see from above. let $t=0$ $\theta=0$

$r = x_0 + v_0 t \qquad \dot{r} = v_0 \qquad \ddot{r} = 0$

$\boxed{\dot{\varphi} = \Omega} \qquad \ddot{\varphi} = 0$

$$\bar{a} = (\cancel{\ddot{r}}^{\,0} - r\dot{\varphi}^2)\,\hat{n}_r + (r\cancel{\ddot{\varphi}}^{\,0} + 2\dot{r}\dot{\varphi})\,\hat{n}_\varphi$$

$$☆ \quad \boxed{= -(x_0+v_0 t)\Omega^2\,\hat{n}_r + 2 v_0 \Omega\,\hat{n}_\varphi}$$

$$(x_0+v_0t)\Omega^2 = (x_0+v_0t)k + \mu m g \quad ?$$

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Newton's Law

1. $\Sigma F = 0 \Rightarrow (v = const\&a = 0)$
2. $\vec{F} = m\vec{a}$
3. $F_{BA} = -F_{AB}$

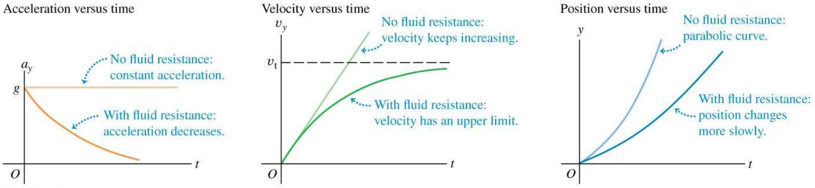
Application to Fluid & Air Drag

For linear air drag, i.e. $mg - kv = ma$

$$a = ge^{-\frac{kt}{m}}$$

$$v = \frac{mg}{k}(1 - e^{-\frac{kt}{m}})$$

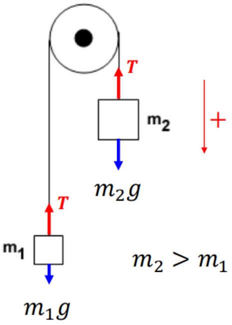
$$y = \frac{mg}{k}t - \frac{m^2 g}{k^2}(1 - e^{-\frac{kt}{m}})$$



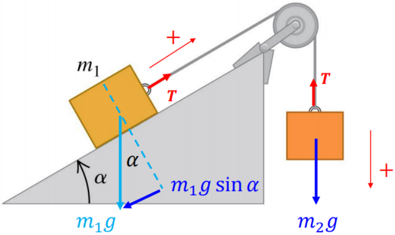
Application to Fluid & Air Drag: General Strategy

1. We have the Newton Law equation $F(v) = ma$
2. Separation of variables
$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F(v)}{m} \Rightarrow \frac{\mathrm{d}v}{F(v)} = \frac{\mathrm{d}t}{m}$$
3. Integration

Frequently Used Model

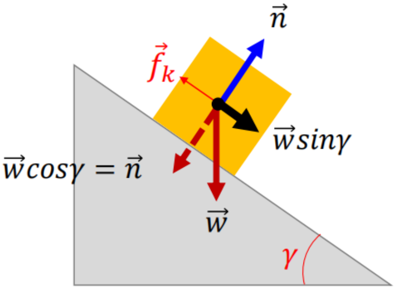


Frequently Used Model



Note: Carefully denote the positive direction, and think about possible negative values.

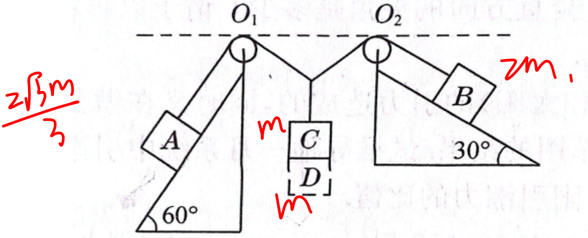
Frequently Used Model



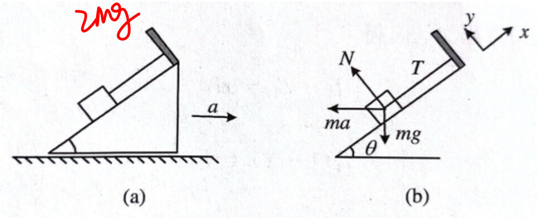
VP150 Mid RC Newton Dynamics Exercise

Zimeng Mao

O_1 *and* O_2 are of the same height and they are frictionless. The mass of A, B and C are $\frac{2\sqrt{3}}{3}m, 2m$ *and* m. The slopes are frictionless. When equilibrium has reached, we put D (mass: m) on C with 0 velocity and CD stick together. What is the acceleration of A and B at the instant we put D?



An object of mass m is placed on a frictionless slope. A light sting connects the object to a plank on the top of the slope. The maximum force the string can stand is 2mg. In order to make sure that m doesn't separate from slope, what is the range of the acceleration of the slope?



速 v x

Write the velocity and displacement formula with regard to time in the following condition: 1) $f = -kv$ 2) $f = -kv$ during the process of a free fall 3) $f = -kv^2$

Mass: m initial velocity: v_0 for 1) & 3)

$$-kv = m\frac{dv}{dt} \quad -\frac{k}{m}dt = \frac{1}{v}dv \quad \Rightarrow -\frac{k}{m}t = \ln v$$

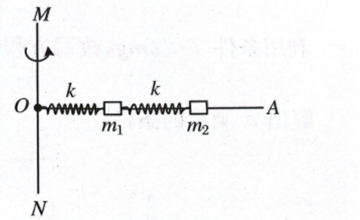
$$v = e^{-\frac{k}{m}t}$$

$$mg - kv = m\frac{dv}{dt}$$

A ship of mass M sails with constant velocity, while its engine provides a constant propelling force of magnitude F_{prop}. There is a linear drag force acting upon the ship, $F_{drag} = -\beta v$, where $\beta > 0$ is constant. At the instant t=0 s, the magnitude of the ship's propelling force suddenly drops to 25% of the original value F_{prop}. Find

- the speed of the ship before the reduction of the propelling force, and
- the dependence of ship's speed on time after the propelling force has been reduced.

A smooth rod OA, can be rotated horizontally around point O, there are two identical springs(k), and two identical objects(m), if the system around the MN axis does stable uniform rotation, find the range of angular velocity of rotation.



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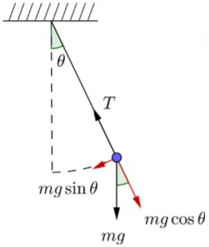
Simple Harmonic Oscillation

Motion Equation

$$\ddot{x} + \omega_0^2 x = 0$$

- Spring: $\ddot{x} + \frac{k}{m}x = 0$
- Simple pendulum: $\ddot{\theta} + \frac{g}{l}\theta = 0$ (Constraint?)



Simple Harmonic Oscillation

General Solution

$$x(t) = A\cos(\omega_0 t + \varphi) = B\sin(\omega_0 t) + C\cos(\omega_0 t)$$

- Natural Frequency: ω_0
- Frequency: $f = \frac{\omega_0}{2\pi}$
- Period: $T = \frac{1}{f} = \frac{2\pi}{\omega_0}$

ω_0 can be directly calculated for a specific oscillator.

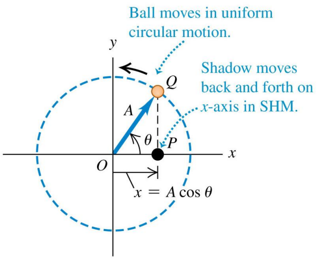
- Spring: $\omega_0 = \sqrt{k/m}$
- Simple Pendulum: $\omega_0 = \sqrt{g/l}$

Simple Harmonic Oscillation

Comparison with Uniform Circular Motion
$\Rightarrow$ Validation of General Solution

$$x = R\cos(\omega t + \varphi) \quad y = R\sin(\omega t + \varphi)$$

$$\begin{cases} a_x = -R\omega^2 \cos(\omega t + \varphi) = -\omega^2 x \\ a_y = -R\omega^2 \sin(\omega t + \varphi) = -\omega^2 y \end{cases}$$



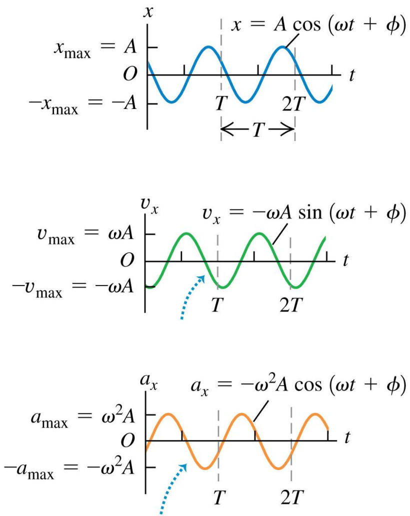
$a_x + \omega^2 x = 0$
$a_y + \omega^2 y = 0$

Simple Harmonic Oscillation: Important Quantity

$$x(t) = A\cos(\omega_0 t + \varphi) \Rightarrow x_{max} = A$$
$$v(t) = -\omega A\sin(\omega_0 t + \varphi) \Rightarrow v_{max} = -\omega A$$
$$a(t) = -\omega^2 A\cos(\omega_0 t + \varphi) \Rightarrow a_{max} = -\omega^2 A$$

- x lags v by $\frac{1}{4}$ cycle ($\frac{pi}{2}, 90^\circ$)
- v lags a by $\frac{1}{4}$ cycle ($\frac{pi}{2}, 90^\circ$)

Simple Harmonic Oscillation: Graph



Damped Oscillation

Motion Equation

$$\ddot{x} + \frac{b}{m}\dot{x} + \omega_0^2 x = 0$$

Classification

- Underdamped Regime: $b < 2m\omega_0$
- Overdamped Regime: $b > 2m\omega_0$
- Critical Damping: $b = 2m\omega_0$

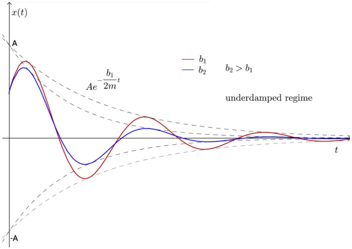
Damped Oscillation: Underdamped $b < 2m\omega_0$

General Solution

$$x(t) = Ae^{-\frac{b}{2m}t}\cos\left(\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}\,t + \varphi\right)$$

- Amplitude: $Ae^{-\frac{b}{2m}t}$
- Angular frequency: $\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$

Damped Oscillation: Underdamped $b < 2m\omega_0$



- Motion is still periodic.
- The amplitude of oscillations decreases exponentially with time ($Ae^{-\frac{b}{2m}t}$).
- Angular frequency $\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} < \omega_0$.
- As b increases, the angular frequency decreases, and the amplitude decreases faster.

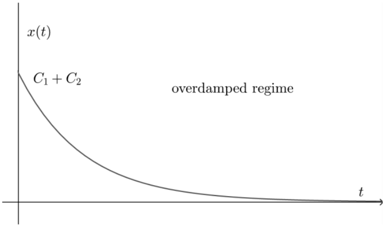
Damped Oscillation: Overdamped $b > 2m\omega_0$

General Solution

$$x(t) = C_1 e^{-\left(\frac{b}{2m} - \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t} + C_2 e^{-\left(\frac{b}{2m} + \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t}$$

Effects of strong damping (overdamped regime)

* No periodic behavior.
* Strong damping results in aperiodic motion: the particle returns aperiodically to the equilibrium position.



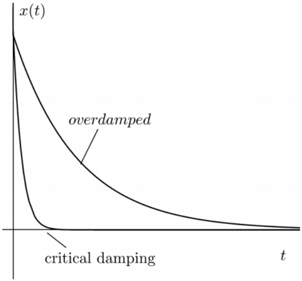
Damped Oscillation: Critical Damping $b = 2m\omega_0$

General Solution

$$x(t) = D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t}$$

Effects of critical damping

* No periodic behavior.
* The system may pass through the equilibrium position at most once (see Problem Set).



Driven Oscillation

Motion Equation

$$\ddot{x} + \frac{b}{m}\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega_{dr} t$$

General Solution

x(t)= $\boxed{\text{solution to the equation of motion for damped oscillator}}$ +
$\boxed{\text{periodic } \textbf{steady-state oscillations} \text{ with angular frequency } \omega_{dr}}$

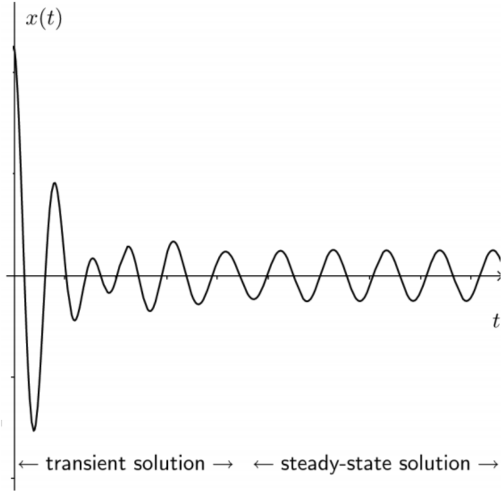
Driven Oscillation

Steady State Solution

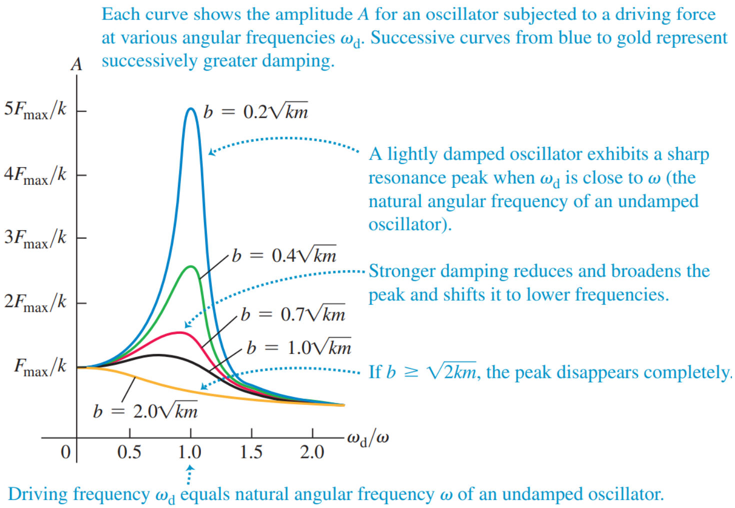
$$x_s(t) = A\cos(\omega_{dr}t + \varphi)$$

- Amplitude: $A(\omega_{dr}) = \dfrac{F_0}{m\sqrt{\left(\omega_0^2 - \omega_{dr}^2\right)^2 + \left(\frac{b\omega_{dr}}{m}\right)^2}}$
- Phase Shift: $\tan\varphi = \dfrac{b\omega_{dr}}{m(\omega_{dr}^2 - \omega_0^2)}$ $\varphi \in [-\pi, 0)$
- Resonance Frequency: $\omega_{res} = \sqrt{\omega_0^2 - \dfrac{b^2}{2m^2}}$

Driven Oscillation: Graph



Driven Oscillation: Mechanical Resonance



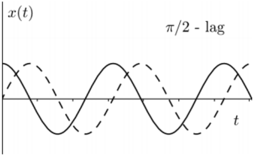
Driven Oscillation: Mechanical Resonance

As we change the driven frequency, the maximum amplitude of the oscillator varies. Resonance is reached when the maximum amplitude is maximum.

- ω_{res} decreases if b increases (stronger drag)
- A_{max} decreases if b increases (stronger drag)

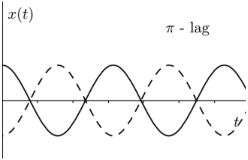
Driven Oscillation: Phase Shift

- If $\omega_{dr} \to \omega_0$ (close to resonance[3]), then $\phi \to -\pi/2$.



The response ($x_s(t)$) lags the drive ($F_{dr}(t)$) by 1/4 of the cycle.

- If $\omega_{dr} \to \infty$ (high frequencies), then $\phi \to -\pi$ the response lags the drive by 1/2 of the cycle (displacement and drive are in antiphase)



The response ($x_s(t)$) lags the drive ($F_{dr}(t)$) by 1/2 of the cycle.

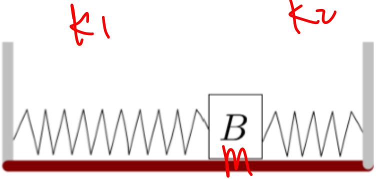
Exercise 1

Consider two harmonic oscillators A and B made of spring–mass systems. The masses attached to the springs are equal in both systems, and both are driven by sinusoidal external forces with the same driving frequency and amplitude. System A oscillates in a liquid with drag coefficient b_A whereas system B in a different liquid with drag coefficient b_B. The relation between the drag coefficients is $b_A < b_B$, yet the resonance frequency for both systems is equal.
What is the relation between the spring constants k_A and k_B of systems A and B, respectively?

Solution

Exercise 2

Two massless springs, obeying Hooke's law with spring constants k_1 and k_2, are used to set up the system shown in the figure. There is no friction in the system and the mass of block B is m. Argue that the system is a **simple harmonic oscillator**. What is the period of oscillations of the block B?



Solution

Exercise 3

An apparatus to determine coefficients of friction is shown in the figure. The spring is non-linear and the magnitude of the force required to keep it stretched or compressed by a length x is equal to Cx^3, where C is a positive constant. At the angle θ shown with the horizontal, the block of mass m just starts to slide on the rough floor of the box. The box then continues to slide a distance d at which point it hits the spring, and compresses the spring a distance x before coming to rest. In terms of the given quantities and fundamental constants, derive an expression for each of the following.

Exercise 3

$\int F\,dx$

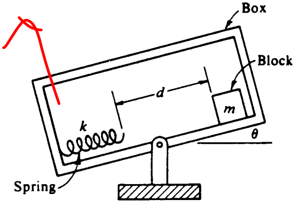
$\int_0^x cx^3\,dx$

$=$

$mg(d+x)\sin\theta$

$-\mu mg(d+x)\cos\theta$

cx^3



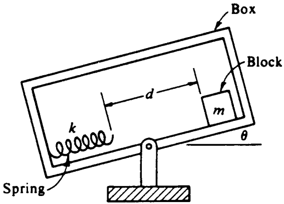
- The coefficient of static friction μ_s between the block and the floor.
- The coefficient of kinetic friction μ_k between the block and the floor.

Exercise 3

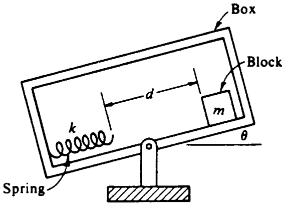
$\ddot{x} + \frac{k}{m} x = 0$

- Assume that the surface of the apparatus, which the block rests on, is perfectly smooth, and that the spring attaches permanently to the block after the block hits it. Argue that the block will move in **periodic motion** and find its amplitude. Is this motion harmonic?

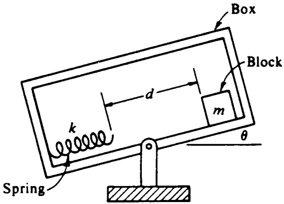
Solution



Solution



Solution



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Reference

- Zhemin Qu, 2023SU VP150 Mid RC.
- Jingjia Peng, 2023SU VP150 Mid RC.
- Mateusz Krzyzosiak, 2024SU VP150 Lec.
- Mesli Abdelmadjid, 2024SU VP150 Lec.